

Lab: The Sandwich Algorithm

Goal

To write a perfect Algorithm (Set of instructions) that a computer could follow.

What You Need

- Bread, Jam/Jelly, Peanut Butter (or pretend items).
- A Knife (Plastic).
- A "Robot" Partner.

Steps

1. The Code (Writing)

Write down the steps to make a PB&J Sandwich. Be detailed!

- Bad Code: "Put PB on bread." (Which bread? Did you open the jar?)
- Good Code: "1. Pick up knife. 2. Scoop PB. 3. Spread on Bread Slice A."

2. The Test (Running)

Read your code to the Robot. They must do *exactly* what you say. If you didn't say "Take the lid off," they will try to scoop through the lid!

3. The Fix (Debugging)

Did it work? If not, change your written steps. Try again until you have a perfect sandwich.

My Code Log

Algorithm Step	Did Robot do it right?
"Put Jam on Bread"	No (Used fingers instead of knife!)
"Use Knife to spread Jam"	Yes!

What We Learned

Computers are dumb! They need humans to tell them exactly what to do. Coding is just writing very clear instructions.